

Nakuja Project

Solid Propulsion Team

N4 One-Grain Static Test — Test Report and Theoretical/Experimental Analysis

Document Version: 1.0 (Post-Execution Report)
Motor Configuration: N4, 1-Grain, Aluminium Casing, BATES Geometry
Date of Execution: Thursday, 16 July 2026
Location: Baseball Pitch
Prepared by: Paschal Peters (Data Recorder / Timer)
Outcome: **SUCCESS** — Full burn achieved, no anomalies flagged

1 Purpose

This report documents the successful live static fire test of the N4 one-grain solid rocket motor (KNSB propellant, aluminium casing, BATES geometry) conducted on 16 July 2026. Unlike previous test days, this attempt proceeded without any faults in the ignition circuit, load cell system, or arming sequence, following the corrective actions applied after the 11 July 2026 MOSFET short-circuit failure. This report goes beyond a procedural log: it presents the theoretical performance profile generated in OpenMotor, the experimental thrust data recorded on test day, a direct comparison between the two, and an analysis of the burn-time discrepancy observed — together with the corrective direction proposed for future grains.

2 Test Overview

Location	Baseball Pitch
Motor	N4, 1-Grain, Aluminium Casing
Procedure Followed	Standard one-grain static test procedure (as previously documented)
Faults During Test	None
Logistics Issue	Delay in transporting the test stand to the baseball pitch, as the project vehicle was unavailable
Outcome	Success — full ignition and burn, motor and instrumentation performed as expected

Although the test itself was executed cleanly, the primary operational challenge of the day was logistical rather than technical: the absence of the project car meant the test stand had to be transported to the baseball pitch by alternative means, which should be planned for in advance of future test days to avoid schedule slippage.

3 Theoretical Performance — OpenMotor Simulation

The N4 grain was modelled in OpenMotor prior to testing using a single BATES-geometry grain with the following configuration:

Parameter	Value
Casing Diameter	86.00 mm
Grain Length	145.00 mm
Core Diameter	35.00 mm
Inhibited Ends	Neither
Nozzle Throat Diameter	23 mm
Propellant	Nakka — KNSB (1)
Propellant Mass	1.23 kg
Port/Throat Ratio	2.32

3.1 Predicted Motor Statistics

Parameter	Value	Parameter	Value
Motor Designation	J352 (58%)	Average Pressure	0.92 MPa
Total Impulse	1014.11 Ns	Peak Pressure	0.95 MPa
Delivered ISP	84.08 s	Initial Kn	61.71
Burn Time	2.85 s	Peak Kn	68.79
Volume Loading	83.44%	Ideal Thrust Coefficient	0.95
Peak Mass Flux	0.48 lb/(in ² s)	Delivered Thrust Coefficient	0.92

The simulation predicts a short, sharply-peaked burn: thrust rises almost immediately after ignition, sustains a broad plateau around 330–380 N, and terminates abruptly just before the 2.85 s mark. This profile reflects the fast, near-neutral regression expected of an uninhibited BATES grain with this port/throat ratio.

4 Experimental Results

The experimental thrust curve was captured for the full duration of the burn and analysed to extract the following key parameters:

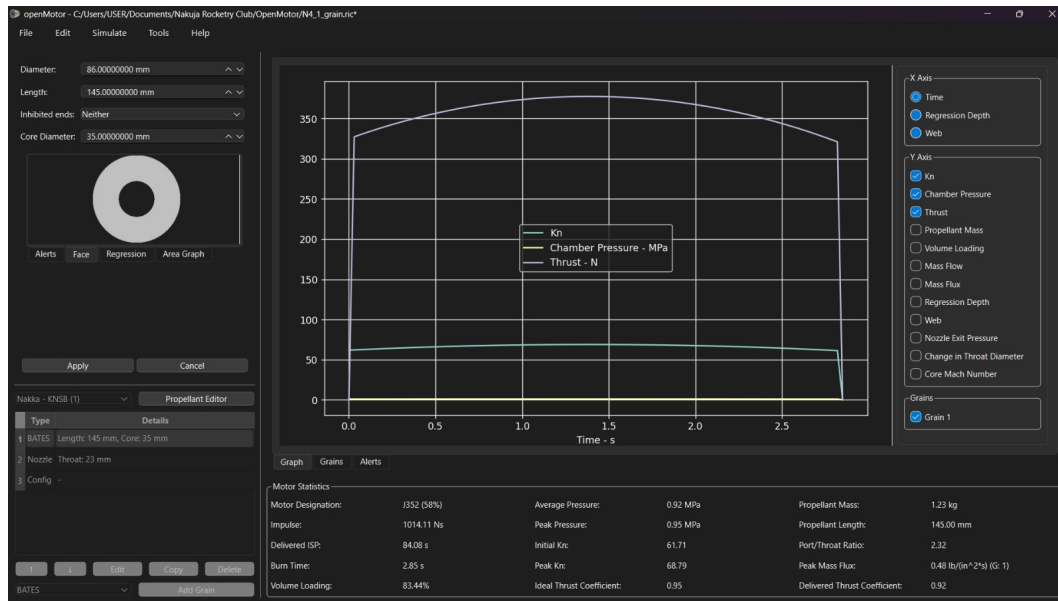


Figure 1: OpenMotor simulation of the N4 1-grain configuration, showing predicted Kn, chamber pressure, and thrust profiles.

Parameter	Value
Peak Thrust	336.33 N
Time of Peak Thrust	8.49 s
Burn Start	0.74 s
Burn End	10.96 s
Burn Time	10.22 s
Total Impulse	2719.01 Ns
Average Thrust (full record)	248.05 N
Average Thrust (during burn)	266.09 N

The experimental curve shows a markedly different shape from the simulation: a comparatively gradual rise over the first ~ 2 seconds, a long mid-burn plateau between roughly 250–295 N lasting several seconds, a late secondary rise to peak thrust at 8.49 s, and a slow taper to burnout at 10.96 s.

5 Theoretical vs. Experimental Comparison

Parameter	Theoretical	Experimental	Deviation
Peak Thrust	~ 378 N	336.33 N	$\approx 11\%$ lower
Burn Time	2.85 s	10.22 s	$\approx 3.6\times$ longer
Total Impulse	1014.11 Ns	2719.01 Ns	$\approx 2.7\times$ higher
Average Thrust (during burn)	≈ 356 N	266.09 N	$\approx 25\%$ lower

Parameter	Theoretical	Experimental	Deviation
Curve Shape	Fast rise, broad neutral plateau, sharp cutoff	Slow rise, mid-burn plateau, late secondary rise, gradual tail	Fundamentally different burn profile

6 Discussion: The Burn Time Discrepancy

The central finding of this test is that while the motor performed successfully — full ignition, sustained thrust, and clean burnout with no hardware faults — the burn duration was approximately **3.6 times longer** than the OpenMotor prediction, and total impulse delivered was correspondingly **2.7 times higher** than predicted, despite a *lower* peak and average thrust.

This pattern is consistent with a grain that is regressing (burning) more slowly and less uniformly than the ideal BATES assumption used in the simulation. The OpenMotor model assumes a clean, fully-bonded, void-free grain surface producing a fast, near-neutral pressure build-up. The experimental curve's extended, irregular plateau instead suggests:

- Slower-than-modelled linear burn rate, possibly due to incomplete propellant densification or residual porosity in the grain.
- Uneven regression across the grain's burning surface, consistent with the microcracking and void defects noted in earlier grain-preparation reviews, which would locally slow or stall combustion instead of allowing the core to regress uniformly.
- Delayed pressure build-up: the theoretical curve reaches peak thrust almost immediately (by ~ 1.3 s) and holds it, while the experimental grain only approaches its peak much later (8.49 s), indicating the chamber is not reaching and sustaining design pressure as quickly as intended.

This reinforces the case, raised in prior grain-preparation reviews, for improving grain homogeneity so that the pressure build-up matches the theoretical curve more closely — burning faster, more uniformly, and closer to the predicted 2.85 s burn window.

7 Recommendation: Surfactant / Primer Addition

To bring the experimental burn profile closer to the theoretical prediction, it is recommended that the grain preparation process incorporate a **surfactant additive (primer)** during casting, alongside the previously adopted vacuum degassing step. The objective is twofold:

- **Reduce burn time** by improving propellant homogeneity and eliminating trapped voids/microcracks that currently slow and prolong regression, bringing burn duration down toward the ~ 2.85 s simulated figure.

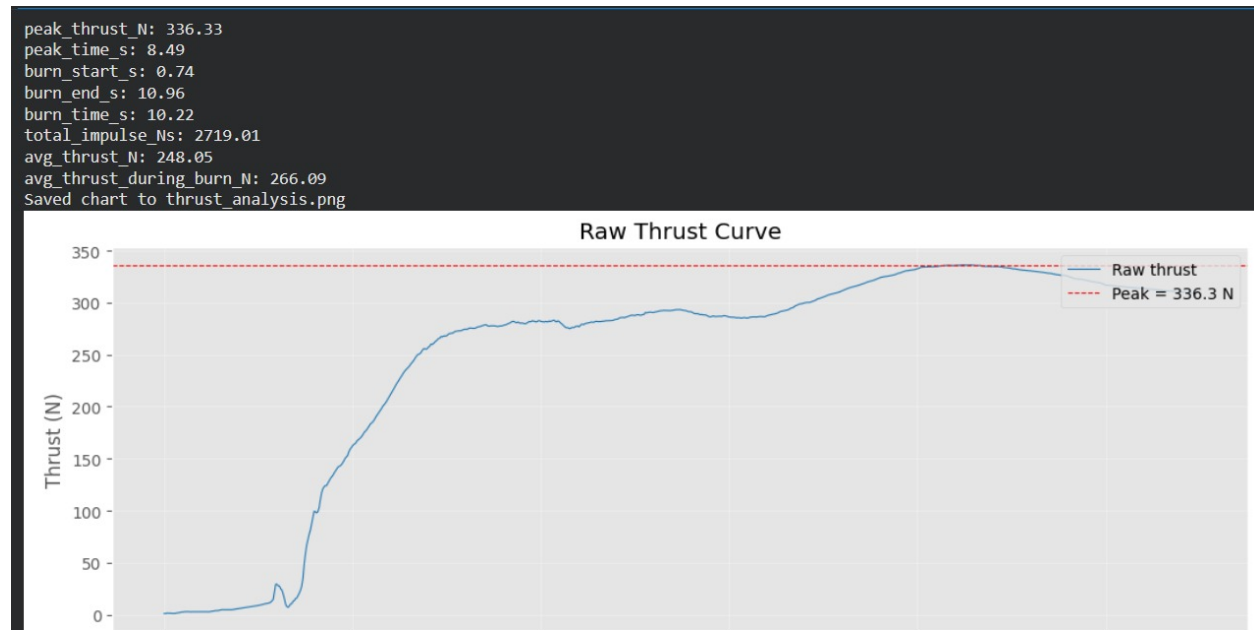


Figure 2: Raw experimental thrust curve with annotated peak thrust (336.3 N).

- **Sharpen pressure build-up** so the chamber reaches and sustains near-peak thrust quickly after ignition, mirroring the fast rise and broad plateau seen in the OpenMotor curve, rather than the gradual, delayed rise currently observed.

A more homogeneous, void-free grain should burn at a rate closer to the modelled linear regression rate, which would simultaneously shorten burn time, raise average thrust (since impulse would be delivered over a shorter interval), and produce a thrust curve shape that more closely tracks the OpenMotor prediction.

8 Conclusion and Next Steps

The 16 July 2026 static test is classified as a **SUCCESS**: the motor ignited and burned fully with no hardware or procedural faults, validating the corrective actions taken after the 11 July MOSFET failure. The key open item is not reliability but **ballistic performance matching** — the experimental burn is significantly longer and less sharply peaked than theory predicts.

- Trial surfactant-treated, vacuum-degassed grains in the next batch and re-run the static test for direct before/after comparison.
- Retain chamber pressure instrumentation (if not already logged) alongside thrust in the next test to directly verify whether pressure build-up rate improves.
- Secure dedicated transport for the test stand ahead of the next test day to avoid recurrence of the logistics delay experienced on 16 July.
- Update the OpenMotor model's assumptions (e.g. propellant burn rate coefficients) if repeated tests continue to show a consistent, reproducible offset from prediction, rather than treating each result as an isolated grain defect.

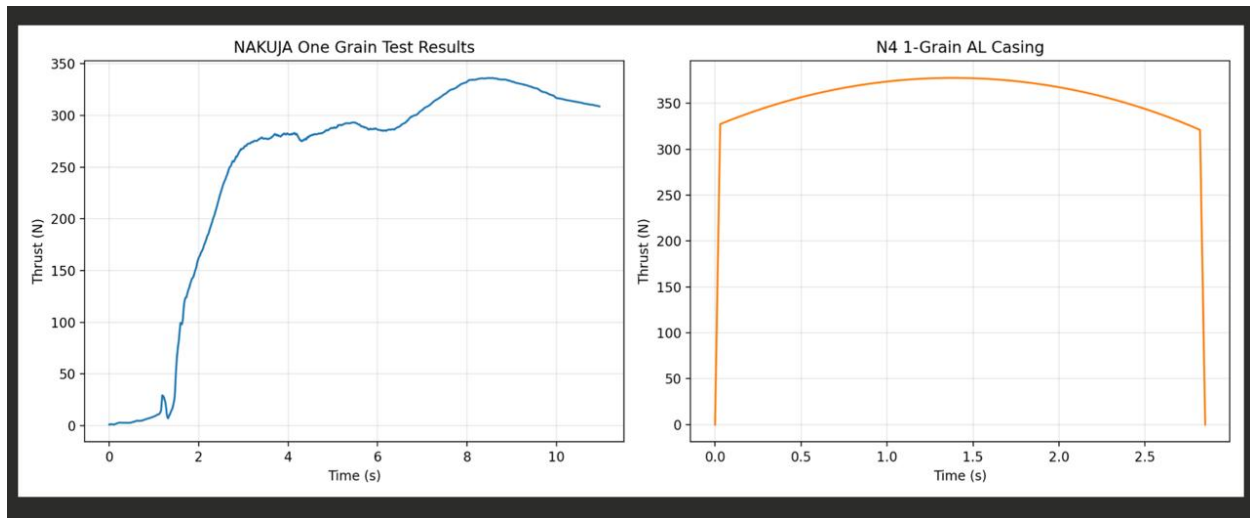


Figure 3: Side-by-side comparison: experimental thrust curve (left) vs. OpenMotor simulated thrust curve (right). Note the difference in both time axis scale and curve shape.

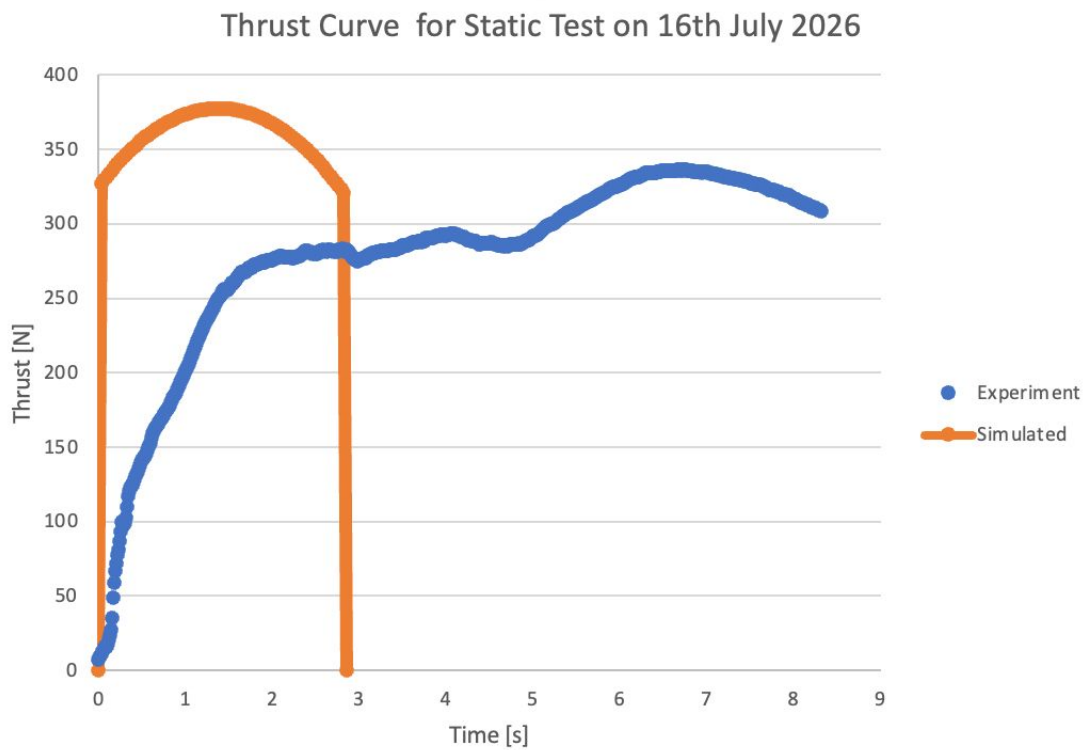


Figure 4: Overlay of experimental (blue) and simulated (orange) thrust curves for the 16 July 2026 static test, plotted on a common time axis.